

VentureMiner AI

Agent Specifications

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Table of Contents

Document 09 — Agent Specifications

Table of Contents

- 1. Purpose & Scope
- 2. Specification template
- 3. Orchestrator (AGT-ORCH)
- 4. Discovery planner (AGT-DISC-PLANNER)
- 5. Discovery clusterer (AGT-DISC-CLUSTER)
- 6. Research — market (AGT-RSRCH-MARKET)
- 7. Research — demand (AGT-RSRCH-DEMAND)
- 8. Research — competitive (AGT-RSRCH-COMP)
- 9. Research — pricing (AGT-RSRCH-PRICING)
- 10. Research — persona (AGT-RSRCH-PERSONA)
- 11. Research — WTP (AGT-RSRCH-WTP)
- 12. Research — GTM (AGT-RSRCH-GTM)
- 13. Research — risk (AGT-RSRCH-RISK)
- 14. Scoring (AGT-SCORE)
- 15. Report writer (AGT-RPT-WRITER)
- 16. Verifier (AGT-VERIFY)
- 17. Safety filter (AGT-SAFETY)
- 18. Critic (AGT-CRITIC)
- 19. Appendix
 - 19.1 Agent ID index
 - 19.2 Revision history
 - 19.3 Cross-references

Document 09 — Agent Specifications

Per-agent contracts. Each agent has a system prompt, a tool set, a state schema, success criteria, and a test plan. This document enumerates them.

Table of Contents

1. Purpose & Scope
2. Specification template
3. Orchestrator (AGT-ORCH)
4. Discovery planner (AGT-DISC-PLANNER)
5. Discovery clusterer (AGT-DISC-CLUSTER)
6. Research — market (AGT-RSRCH-MARKET)
7. Research — demand (AGT-RSRCH-DEMAND)
8. Research — competitive (AGT-RSRCH-COMP)
9. Research — pricing (AGT-RSRCH-PRICING)
10. Research — persona (AGT-RSRCH-PERSONA)
11. Research — WTP (AGT-RSRCH-WTP)
12. Research — GTM (AGT-RSRCH-GTM)
13. Research — risk (AGT-RSRCH-RISK)
14. Scoring (AGT-SCORE)
15. Report writer (AGT-RPT-WRITER)
16. Verifier (AGT-VERIFY)
17. Safety filter (AGT-SAFETY)
18. Critic (AGT-CRITIC)
19. Appendix

1. Purpose & Scope

This document is the per-agent contract. Each agent must have:

- A unique ID and name.
- A purpose (one sentence).
- A system prompt (kept current in code; summarized here).
- A typed input/output schema.
- A tool set.
- Failure modes.
- Success criteria and test plan.

```

agent:
id: AGT-
name:
purpose:
inputs:
-
outputs:
-
tools:
-
dependencies:
- rag-svc | memory-svc | plugin-svc | ...
failure_modes:
-
success_criteria:
-
test_plan:
-

```

The full per-agent YAMLS live in agent-runtime/agents//contract.yaml. This document is the index.

3. Orchestrator (AGT-ORCH)

- Purpose: Plan a run, dispatch specialists, manage state and budget.
- Inputs: RunRequest{goal, depth, workspace_id, user_id}.
- Outputs: final RunState with evidence, scores, report populated.
- Tools: none directly; dispatches specialists via in-process or NATS calls.
- Dependencies: RAG (read), memory (read/write), all specialists.
- Failure modes: budget exceeded, planner failure, unresolvable conflict.
- Success criteria: a complete run with all dimensions scored and cited.
- Test plan: mini-graph replays, end-to-end replays, adversarial planners.

4. Discovery planner (AGT-DISC-PLANNER)

- Purpose: Given a seed and a window, design a discovery run (sources, queries, depth).
- Inputs: seed, window, depth, source list.
- Outputs: DiscoveryPlan{sources, queries, expected_yield}.
- Tools: source metadata; planner-only tools (no external calls).
- Success criteria: plan that, when executed, produces $\geq$ expected yield with cited hits.
- Test plan: replay on 30 fixed seeds; expected yield threshold.

- Inputs: list[DiscoveryHit].
- Outputs: list[CandidateOpportunity{title, summary, source_ids}].
- Tools: none external; uses embedding-based clustering.
- Success criteria: intra-cluster similarity > 0.85; inter-cluster similarity < 0.6.
- Test plan: synthetic clusters; cluster purity checks.

6. Research — market (AGT-RSRCH-MARKET)

- Purpose: Estimate TAM/SAM/SOM and growth for an opportunity.
- Inputs: Opportunity, source list, RAG.
- Outputs: MarketEstimate{tam_usd, sam_usd, som_usd, growth_yoy_pct, sources, confidence}.
- Tools: market data plugins (Statista-like, free data); RAG.
- Success criteria: estimate has ≥ 3 sources; ranges given; confidence assigned.
- Test plan: known markets replay; expert-set tolerance band.

7. Research — demand (AGT-RSRCH-DEMAND)

- Purpose: Aggregate demand signals (search volume, social volume, intent).
- Inputs: Opportunity, source list, RAG.
- Outputs: DemandSignals{search_volume, social_volume, intent_signals, sources}.
- Tools: Google Trends plugin, social APIs, RAG.
- Success criteria: at least 2 independent demand sources; signal strength graded.
- Test plan: replay on known keywords.

8. Research — competitive (AGT-RSRCH-COMP)

- Purpose: Build a competitive map (competitors, positioning, gaps).
- Inputs: Opportunity, source list, RAG.
- Outputs: CompetitorMap{competitors:[{name, positioning, features, pricing_band}], gaps:[...]}
- Tools: web search, RAG, app store data, G2.
- Success criteria: ≥ 3 named competitors; each cited; ≥ 1 gap identified.
- Test plan: replay on known competitive markets.

9. Research — pricing (AGT-RSRCH-PRICING)

- Purpose: Collect competitor pricing tiers and ranges.
- Inputs: Opportunity, CompetitorMap.
- Outputs: PricingBenchmark{tiers:[{competitor, plan, price, currency, as_of}]}

- Test plan: replay on known products.

10. Research — persona (AGT-RSRCH-PERSONA)

- Purpose: Synthesize 1–3 buyer personas from evidence.
- Inputs: Opportunity, CompetitorMap, RAG.
- Outputs: Personas:[{name, demographics, pains, jobs, sources}].
- Tools: RAG, web search.
- Success criteria: each persona has ≥ 3 sourced pains/jobs.
- Test plan: replay on known audiences.

11. Research — WTP (AGT-RSRCH-WTP)

- Purpose: Estimate willingness to pay.
- Inputs: Opportunity, PricingBenchmark, Personas, RAG.
- Outputs: WTP{low, mid, high, currency, rationale, sources}.
- Tools: RAG, web search.
- Success criteria: range is plausible vs. competitor pricing; rationale cited.
- Test plan: replay on known markets.

12. Research — GTM (AGT-RSRCH-GTM)

- Purpose: Diagnose top 3 GTM channels used by competitors + estimated CAC.
- Inputs: Opportunity, CompetitorMap, RAG.
- Outputs: GTM{channels:[{name, evidence, cac_band}]}.
- Tools: web search, RAG.
- Success criteria: each channel has ≥ 1 cited source; CAC band present.
- Test plan: replay on known categories.

13. Research — risk (AGT-RSRCH-RISK)

- Purpose: Identify top risks and mitigations.
- Inputs: Opportunity, all prior research.
- Outputs: RiskRegister{risks:[{name, likelihood, impact, mitigation, sources}]}.
- Tools: RAG.
- Success criteria: ≥ 5 risks; each with mitigation; no generic boilerplate.
- Test plan: replay; reviewer scores specificity.

- Inputs: evidence, rubric_version.
- Outputs: Score{total, breakdown, rationale, confidence}.
- Tools: none (pure compute + LLM judge).
- Success criteria: score total is within ± 0.5 of expert on a calibration set.
- Test plan: monthly calibration; rubric sensitivity.

15. Report writer (AGT-RPT-WRITER)

- Purpose: Assemble a long-form report from evidence + scores.
- Inputs: evidence, scores, rubric, persona context, report template.
- Outputs: Report{sections:[{title, body, citations}]}
- Tools: RAG, chart render.
- Success criteria: every claim cited; report fits 10–25 pages; 95% acceptance.
- Test plan: golden set; report grading rubric; operator acceptance.

16. Verifier (AGT-VERIFY)

- Purpose: Audit another agent's output for citation, consistency, policy.
- Inputs: candidate output + source corpus.
- Outputs: Verdict{ok, issues:[{severity, message, evidence}]}
- Tools: RAG, policy service.
- Success criteria: recall ≥ 0.9 on planted issues.
- Test plan: planted-issue test set; weekly recalibration.

17. Safety filter (AGT-SAFETY)

- Purpose: Redact PII; enforce content policy.
- Inputs: text, call context.
- Outputs: redacted text + flags.
- Tools: PII detector, policy lookup.
- Success criteria: $\geq 99\%$ PII recall on test set; zero policy violations.
- Test plan: planted-PII test set; red-team review.

18. Critic (AGT-CRITIC)

- Purpose: Review a draft report and propose specific revisions.
- Inputs: draft report, evidence, persona.
- Outputs: Revisions:[{section, change, rationale}]

- Test plan: paired report comparison.

19. Appendix

19.1 Agent ID index

AGT-ORCH	Orchestrator
AGT-DISC-PLANNER	Discovery planner
AGT-DISC-CLUSTER	Discovery clusterer
AGT-RSRCH-MARKET	Research — market
AGT-RSRCH-DEMAND	Research — demand
AGT-RSRCH-COMP	Research — competitive
AGT-RSRCH-PRICING	Research — pricing
AGT-RSRCH-PERSONA	Research — persona
AGT-RSRCH-WTP	Research — WTP
AGT-RSRCH-GTM	Research — GTM
AGT-RSRCH-RISK	Research — risk
AGT-SCORE	Scoring
AGT-RPT-WRITER	Report writer
AGT-VERIFY	Verifier
AGT-SAFETY	Safety filter
AGT-CRITIC	Critic

19.2 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

19.3 Cross-references

- Multi-Agent: Document 08.
- System Architecture: Document 07.
- Scoring Engine: Document 16.
- Report Generation: Document 17.

End of Document 09 — Agent Specifications. Per-agent YAML contracts are stored in `agent_runtime/agents//contract.yaml` and treated as code.